AUTOMATED TEST REPORT

Multi-Channel Sensor Data Analysis | Test ID: TST-2026-001

Component: Rotating Component — Test Article 01

1. Test Configuration

Test ID:	TST-2026-001	**Date/Time:**	2026-03-22 11:14
Component:	Rotating Component — Test Article 01	**Sampling Rate:**	10,000 Hz
Filter Cutoff:	2000 Hz	**Filter Type:**	Butterworth order 4
Channels:	4 (Vibration, Force, Torque, Temperature)	**Duration:**	5.0 s

2. Overall Test Result

■ **PASS**

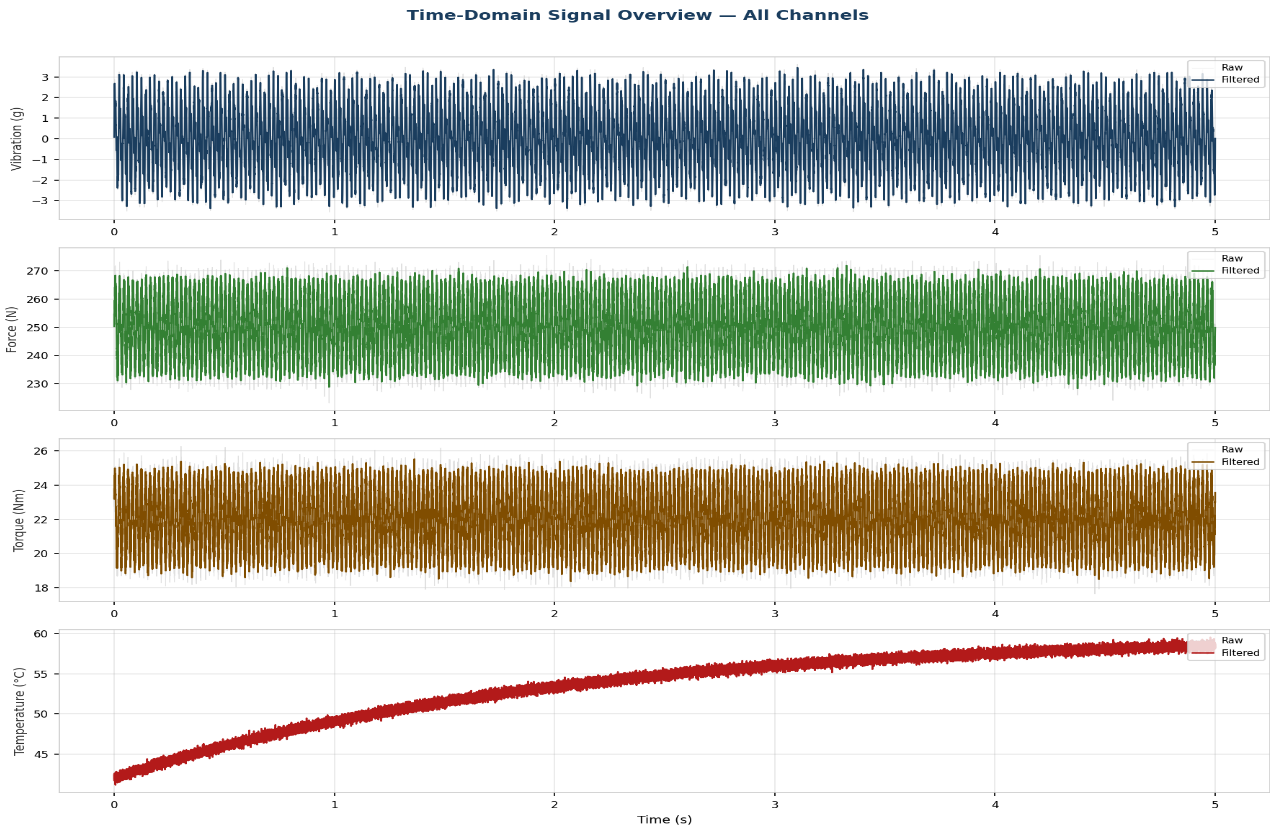
3. Channel Metrics vs. Acceptance Limits

Channel	Metric	Measured	Limit	Result
Vibration	RMS (g)	1.6020	5.0	✓ PASS
Force	Peak (N)	271.95	500.0	✓ PASS
Torque	RMS (Nm)	22.0745	50.0	✓ PASS
Temperature	Max (°C)	59.56	85.0	✓ PASS

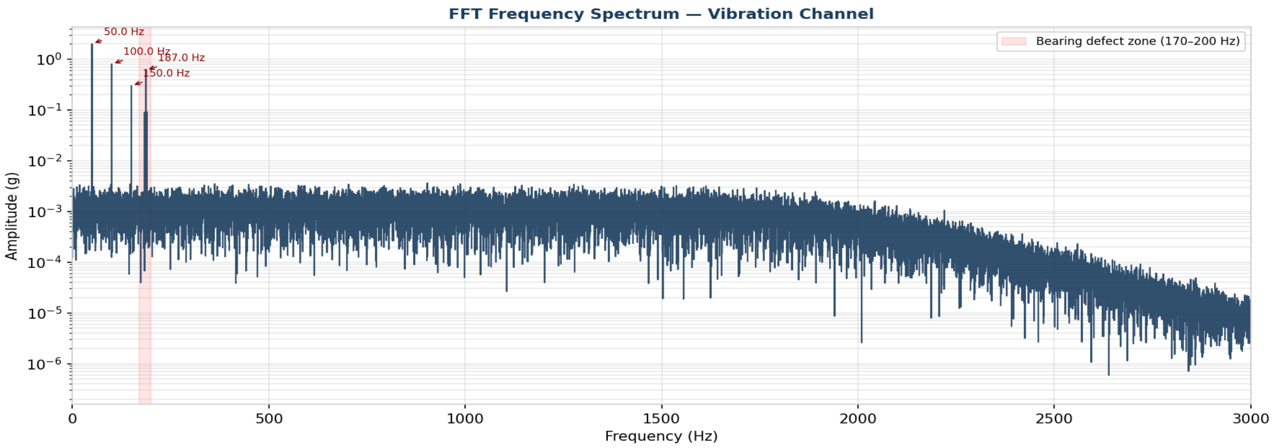
4. FFT Analysis — Dominant Frequency Peaks

Rank	Frequency (Hz)	Amplitude (g)	Assessment
1	50.0	2.0013	Expected (RPM harmonic)
2	100.0	0.8012	Expected (RPM harmonic)
3	187.0	0.6028	■ Investigate — unexpected frequency
4	150.0	0.2992	Expected (RPM harmonic)

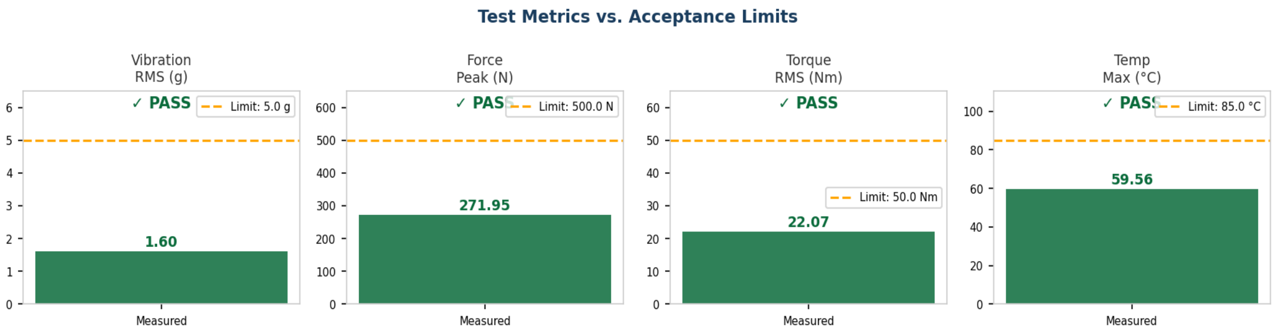
5. Time-Domain Signal Overview



6. FFT Frequency Spectrum — Vibration Channel



7. Metrics vs. Acceptance Limits



8. Conclusions & Recommendations

Channels within acceptance limits: Vibration RMS, Force Peak, Torque RMS, Temperature.

■ FFT analysis identified an unexpected frequency peak at approximately 187 Hz in the vibration channel. This frequency does not correspond to any expected RPM harmonic (50, 100, 150 Hz). It is consistent with a bearing defect frequency signature. Recommendation: inspect bearing condition before next test run. Repeat test after inspection to confirm resolution.

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